## PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

## ARTICLE DETAILS

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| **TITLE (PROVISIONAL)** | Use of tyrosine kinase inhibitors for pediatric philadelphia chromosome-positive acute lymphoblastic leukemia: a systematic review and meta-analysis. |
| **AUTHORS** | Chen, Min; zhu, yiping; Lin, Yunzhu; Tengwang, Tianzi; zhang, lingli |

## VERSION 1 – REVIEW

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| **REVIEWER** | Shouhao Zhou<br>Penn State University, USA |
| **REVIEW RETURNED** | 09-Aug-2020 |

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| **GENERAL COMMENTS** | This is a study of systematic review and meta-analysis to examine effectiveness and safety of TKIs in the management of pediatric Ph+ ALL. My reivew and the following comments are with a particular emphasis on the study design and data analyses.<br><br>1. In general, the principle / standard procedure of meta-analysis were followed properly.<br><br>2. The inclusion criteria identified 2 RCTs and four cohort studies. The total sample size of 536 is small. Out of 2 RCTs, one was between 2 TKIs without a non-TKI-control arm, which is appropriate for the objective of this study. Given the limitation and defect in the data, the clinical impact of the manuscript is very limited. |

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| **REVIEWER** | Francesca Carofiglio<br>Università degli studi di Bari, Aldo Moro, Italy. |
| **REVIEW RETURNED** | 17-Aug-2020 |

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| **GENERAL COMMENTS** | The manuscript is well written on the whole, but it presents some grammatical and syntactic errors (improper use of tenses, plurals and punctuation, use of substantives instead of adjectives). It would be more appreciated if the figures were incorporated into the text. I suggest to check all the the abbreviations. The topic discussed is well introducted and the results are coherent with the conclusion. |

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| **REVIEWER** | Charles Stiller<br>National Cancer Registration and Analysis Service, Public Health England, UK |
| **REVIEW RETURNED** | 20-Aug-2020 |

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| **GENERAL COMMENTS** | The advent of tyrosine kinase inhibitors (TKIs) revolutionised treatment and outcome for Philadelphia chromosome positive |

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| | (Ph+) leukaemias. The authors have carried out the first systematic review and meta-analysis of outcomes and safety of TKIs in paediatric Ph+ acute lymphoblastic leukaemia. They have used appropriate, standard methodology for searching, study selection, bias assessment, analysis, and presentation of results. In general, these methods have been applied correctly, but I do have some specific questions and reservations about the manuscript.<br>1. The rarity of Ph+ ALL in this age group and the dramatic improvement in survival with TKIs, however, inevitably mean that data other than from uncontrolled studies are very scarce. There is only one available study comparing imatinib and dasatinib (Shen, 2020). It therefore seems superfluous to perform a meta-analysis of imatinib versus dasatinib when it amounts to no more than restating the results of the single study. Would it not be better to retain just the quality assessment of that study in the Results, and then state in the Discussion that there is as yet only one study comparing outcomes or safety between these two TKIs and summarise its findings?<br>2. The Methods section should state whether results in the RCTs were compared with patient groups defined by intention to treat or "as treated", and the choice should be justified. Results for both classifications are included in the main paper reporting EsPhALL2004 (Biondi, 2012). They should also be available for the most recent trial (Shen, 2020) either in the published paper or from the investigators.<br>3. A fundamental consideration for systematic review and meta-analysis is that as far as possible the case series should be independent, with overlap and consequent double-counting eliminated. From the information in Table 1 and lists of authors for the references, there is clearly considerable overlap between references 24 (Guo, 2015) and 25 (Liu, 2013). While this does not matter for overall survival (results only in reference 25) and adverse drug reactions (results only in reference 24), event-free survival is reported in both studies. Because reference 25 (Liu, 2013) covers a longer period with larger number of patients, its results for event-free survival should be included but those from reference4 24 (Guo, 2015) should be excluded.<br>4. For EsPhALL2004, would it be better to use the updated results (Biondi, 2019) where possible rather than those in the original publication (Biondi, 2012)? |

**VERSION 1 – AUTHOR RESPONSE**

Reviewers' Comments to Author:
Reviewer: 1
Reviewer Name: Shouhao Zhou
Institution and Country: Penn State University, USA
Please state any competing interests or state 'None declared': None declared

This is a study of systematic review and meta-analysis to examine effectiveness and safety of TKIs in the management of pediatric Ph+ ALL. My reivew and the following comments are with a particular emphasis on the study design and data analyses.

1. In general, the principle / standard procedure of meta-analysis were followed properly.
Response: Many thanks for these comments!

2. The inclusion criteria identified 2 RCTs and four cohort studies. The total sample size of 536 is small. Out of 2 RCTs, one was between 2 TKIs without a non-TKI-control arm, which is appropriate for the objective of this study. Given the limitation and defect in the data, the clinical impact of the manuscript is very limited.
Response: Many thanks for these useful and valid suggestions! Our systematic review included both prospective and retrospective studis and evidence quality was assessed by the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) approach. Although the sample size was small, the present review illustrated the best evidence to assess the effectiveness and safety of TKIs in the management of pediatric Ph+ ALL. The research results can provide references for healthcare providers, consumers, researchers and policy makers to make healthcare decisions.

Reviewer: 2
Reviewer Name: Francesca Carofiglio
Institution and Country: Università degli studi di Bari, Aldo Moro, Italy.
Please state any competing interests or state 'None declared': None declared.

The manuscript is well written on the whole, but it presents some grammatical and syntactic errors (improper use of tenses, plurals and punctuation, use of substantives instead of adjectives). It would be more appreciated if the figures were incorporated into the text. I suggest to check all the the abbreviations. The topic discussed is well introducted and the results are coherent with the conclusion.
Response: Many thanks for these useful and valid suggestions! We have checked all the the abbreviations. The manuscript has been further revised by a native English-speaker to improve the quality of English. The figures need to be uploaded separately and not merged into the text.

Reviewer: 3
Reviewer Name: Charles Stiller
Institution and Country: National Cancer Registration and Analysis Service, Public Health England, UK
Please state any competing interests or state 'None declared': None declared

The advent of tyrosine kinase inhibitors (TKIs) revolutionised treatment and outcome for Philadelphia chromosome positive (Ph+) leukaemias. The authors have carried out the first systematic review and meta-analysis of outcomes and safety of TKIs in paediatric Ph+ acute lymphoblastic leukaemia. They have used appropriate, standard methodology for searching, study selection, bias assessment, analysis, and presentation of results. In general, these methods have been applied correctly, but I do have some specific questions and reservations about the manuscript.
Response: Many thanks for these comments!

1. The rarity of Ph+ ALL in this age group and the dramatic improvement in survival with TKIs, however, inevitably mean that data other than from uncontrolled studies are very scarce. There is only one available study comparing imatinib and dasatinib (Shen, 2020). It therefore seems superfluous to perform a meta-analysis of imatinib versus dasatinib when it amounts to no more than restating the results of the single study. Would it not be better to retain just the quality assessment of that study in the Results, and then state in the Discussion that there is as yet only one study comparing outcomes or safety between these two TKIs and summarise its findings?
Response: Many thanks for these useful and valid suggestions! We have adjusted the structure of the Results section into two parts: "TKI versus non-TKI" and "Different TKIs" . In the "Different TKIs" part, we described the results of the included study (Shen 2020) and eliminated the forest figures of

meta-analysis of the effectiveness of TKIs on overall survival, event-free survival and adverse drug reaction.

2. The Methods section should state whether results in the RCTs were compared with patient groups defined by intention to treat or "as treated", and the choice should be justified. Results for both classifications are included in the main paper reporting EsPhALL2004 (Biondi, 2012). They should also be available for the most recent trial (Shen, 2020) either in the published paper or from the investigators.
Response: Many thanks for these useful and valid suggestions! In order to reduce bias and obtain more reliable results, the intention-to-treat results were used for meta-analysis. In the Methods section, we added "If different analysis methods (such as intention-to-treat analysis, as-treated analysis) are used in the evaluation of benefits between groups, we adopt the results of the intention-to-treat analysis to reduce bias."

3. A fundamental consideration for systematic review and meta-analysis is that as far as possible the case series should be independent, with overlap and consequent double-counting eliminated. From the information in Table 1 and lists of authors for the references, there is clearly considerable overlap between references 24 (Guo, 2015) and 25 (Liu, 2013). While this does not matter for overall survival (results only in reference 25) and adverse drug reactions (results only in reference 24), event-free survival is reported in both studies. Because reference 25 (Liu, 2013) covers a longer period with larger number of patients, its results for event-free survival should be included but those from reference4 24 (Guo, 2015) should be excluded.
Response: Many thanks for these useful and valid suggestions! In the "Characteristics of Included Studies", we added "Two of these studies came from the same research institution. Among them, one study 25 retrospectively analyzed 53 cases between 2008 to 2013. The other 26 retrospectively analyzed 92 cases between 2003 to 2012." In the Results section, we added "Because Liu et al covered a longer period with larger number of patients, its results for EFS was included and those from Guo et al was excluded".

4. For EsPhALL2004, would it be better to use the updated results (Biondi, 2019) where possible rather than those in the original publication (Biondi, 2012)?
Response: Many thanks for these useful and valid suggestions! The results of the long-term follow up of the EsPhALL2004 study (Biondi, 2019) showed that event-free survival (EFS) was stable after about 4 years. The outcome in good risk patients by intention to treat showed a 5-year event-free survival (DFS) in the imatinib arm of 75.5% (SE 6.4) versus 61.4% (SE 7.3) in the no imatinib arm (*P*=0.20). These data confirmed that the introduction of imatinib decreased the rate of relapses and conveyed a marked improvement in EFS. However, the difference was not significant. Due to the lack of survival curves of overall survival and EFS between comparative groups, we still used the data of 3 years of follow-up from EsPhALL2004 study for analysis.

5. The whole manuscript should be checked for idiomatic English language.
Response: Many thanks for these useful and valid suggestions! The manuscript has been further revised by a native English-speaker and checked carefully to improve the quality of English.

## VERSION 2 – REVIEW

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| **REVIEWER** | Charles Stiller<br>National Cancer Registration and Analysis Service, Public Health England, UK |
| **REVIEW RETURNED** | 17-Oct-2020 |

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| **GENERAL COMMENTS** | All review comments on the previous version have been satisfactorily addressed in this resubmssion. |